

Quasi-Static Limits & Doppler Spectrum at 26 GHz

ELEC-H415 (Chapter 4)

Lucas Placentino

École Polytechnique de Bruxelles
Université libre de Bruxelles

2026

Physical Context: The Quasi-Static Hypothesis at 26 GHz

The Quasi-Static Approximation (Course Definition)

In telecommunication systems, the channel is assumed **time-invariant** during the transmission of a block of symbols of duration t_s :

$$y(t) = h(t)x(t) \xrightarrow{t_s \ll \Delta t_c} y(t) \simeq h \cdot x(t)$$

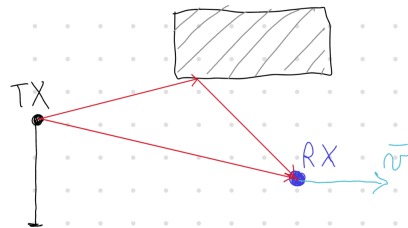
We speak about a **slow fading channel**.

The mmWave Physical Challenge

At carrier frequency $f_c = 26$ GHz:

$$\lambda = \frac{c}{f_c} = \frac{3 \times 10^8 \text{ m/s}}{26 \times 10^9 \text{ Hz}} \approx 1.154 \text{ cm}$$

Because λ is on the centimeter scale, does the quasi-static assumption remain valid for a moving receiver?



Wavelength $\lambda = 1.15$ cm
 $\Rightarrow$ Millimeter-scale phase shifts

Local Area Formulation

Definition (Local Area)

A region of space where the received signal is made of a given set of N plane waves of **constant amplitudes** a_n .

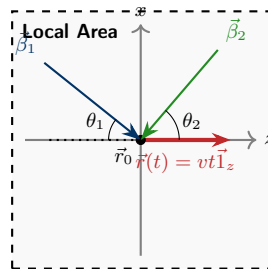
Let the receiver move along trajectory $\vec{r}(t)$. At the origin ($\vec{r} = \vec{0}$), each Multipath Component (MPC) n has delay τ_{n0} and phase ϕ_{n0} . When moving, its phase changes as a plane wave:

$$e^{j\phi_n(t)} e^{-j2\pi f_c \tau_n} = \underbrace{e^{j\phi_{n0}} e^{-j2\pi f_c \tau_{n0}}}_{\triangleq e^{j\varphi_n}} e^{-j\vec{\beta}_n \cdot \vec{r}(t)}$$

where $\varphi_n = \phi_{n0} - 2\pi f_c \tau_{n0}$.

Local Area Channel Transfer Function

$$h(t) = \sum_{n=1}^N a_n e^{j\varphi_n} e^{-j\vec{\beta}_n \cdot \vec{r}(t)}$$



Step-by-Step Derivation of the Doppler Shift

Let the receiver move at constant speed v along the z -axis: $\vec{r}(t) = vt\vec{1}_z$. Consider a single plane wave ($N = 1$) with amplitude $a = 1$ arriving at angle θ relative to the z -axis:

1. Wavevector Definition:

$$\vec{\beta} = -\beta \cos \theta \vec{1}_z - \beta \sin \theta \vec{1}_x$$

where $\beta = \frac{\omega_c}{c} = \frac{2\pi}{\lambda}$ is the wavenumber.

2. Spatial Inner Product:

$$\vec{\beta} \cdot \vec{r}(t) = (-\beta \cos \theta \vec{1}_z) \cdot (vt \vec{1}_z) = -\beta vt \cos \theta$$

3. Channel Transfer Function:

$$h(t) = e^{-j\vec{\beta} \cdot \vec{r}(t)} = e^{j\beta v \cos \theta \cdot t} = e^{j\omega_D t}$$

Doppler Angular Frequency (ω_D)

$$\omega_D \triangleq \beta v \cos \theta = \frac{2\pi}{\lambda} v \cos \theta$$

Maximal Doppler Shift (f_m)

Obtained for head-on incidence
($\theta = 0 \implies \cos \theta = 1$):

$$f_m \triangleq \frac{\omega_m}{2\pi} = \frac{\beta v}{2\pi} = \frac{v}{\lambda} = \frac{v f_c}{c}$$

The Doppler frequency shift is bounded by:

$$f_D \in [-f_m, +f_m]$$

Physical Interpretation: Continuous Wave (CW) Transmission

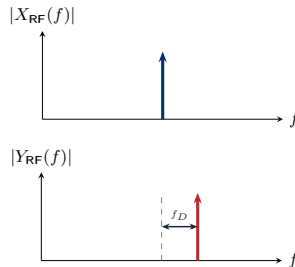
Consider an unmodulated unit-amplitude carrier transmission: $x(t) = 1$.

Baseband Output:

$$y(t) = h(t)x(t) = h(t) = e^{j\omega_D t}$$

Physical Real RF Received Signal:

$$\begin{aligned} y_{\text{RF}}(t) &= \text{Re} \left\{ y(t)e^{j\omega_c t} \right\} = \text{Re} \left\{ e^{j\omega_D t} e^{j\omega_c t} \right\} \\ &= \text{Re} \left\{ e^{j(\omega_c + \omega_D)t} \right\} \\ &= \boxed{\cos \left((\omega_c + \omega_D)t \right)} \end{aligned}$$



Physical Interpretation

The carrier frequency is shifted from f_c to $f_c + f_D$. The channel acts as a frequency modulator through receiver velocity.

The Doppler Spectrum for N Multipath Components

When N multipaths arrive simultaneously from angles θ_n :

$$h(t) = \sum_{n=1}^N a_n e^{j\varphi_n} e^{j\omega_{Dn}t} \quad \text{with} \quad \omega_{Dn} = \beta v \cos \theta_n = 2\pi f_m \cos \theta_n$$

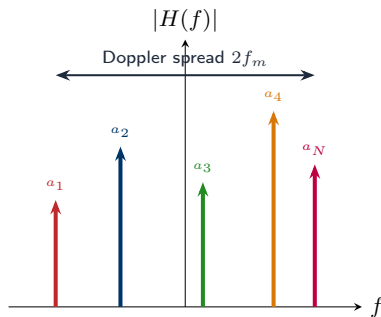
Spectral Representation of $h(t)$

$$H(f) = \mathcal{F}\{h(t)\} = \sum_{n=1}^N a_n e^{j\varphi_n} \delta(f - f_{Dn})$$

The received spectrum is spread over a finite bandwidth called the **Doppler Spread**:

$$B_D = 2f_m = \frac{2v}{\lambda} = \frac{2vf_c}{c}$$

This Doppler spectrum characterizes the **rate of time variation** of the channel alone.



Channel Coherence Time (Δt_c)

Derivation from Spectral Spread

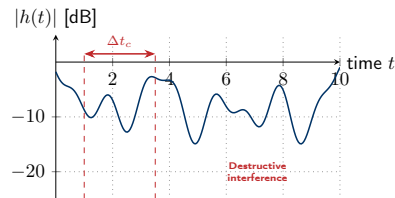
As with any function of time, the time-scale of variation of $h(t)$ is inversely proportional to its spectral width ($2f_m$):

$$\Delta t_c \simeq \frac{1}{2f_m} = \frac{1}{2} \frac{\lambda}{v}$$

Condition for Quasi-Static Fading

Let t_s be the duration of a transmitted block of symbols:

- If $t_s \ll \Delta t_c$: The channel is constant over the block $\Rightarrow$ **Slow Fading** (quasi-static approximation holds).
- If $t_s \sim \Delta t_c$ or $t_s > \Delta t_c$: The channel varies during transmission $\Rightarrow$ **Fast Fading** (quasi-static approximation fails).



Interference pattern causes fast fading in time as the receiver moves.

Spatial Coherence Distance (Δd_c) & Standing Waves

Coherence Distance Definition: The spatial distance traveled by the receiver at speed v during the coherence time Δt_c :

$$\Delta d_c = v \cdot \Delta t_c = v \left(\frac{1}{2} \frac{\lambda}{v} \right) = \frac{\lambda}{2}$$

Proof via 2-Ray Standing Wave

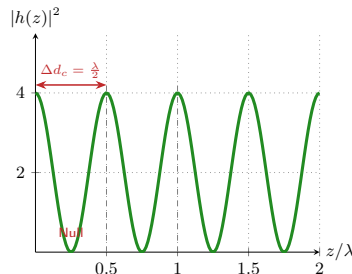
Two equal-amplitude waves arriving in opposite directions along z ($\theta_1 = 0$, $\theta_2 = \pi$):

$$h(z) = e^{-j\beta z} + e^{+j\beta z} = 2 \cos(\beta z)$$

The power pattern is:

$$|h(z)|^2 = 4 \cos^2 \left(\frac{2\pi}{\lambda} z \right) = 2 \left[1 + \cos \left(\frac{4\pi}{\lambda} z \right) \right]$$

The spatial power period is $\frac{\lambda}{2}$. Peak to destructive null is $\frac{\lambda}{4}$.



Spatial scale at 26 GHz:

$$\Delta d_c = \frac{11.54 \text{ mm}}{2} = \mathbf{5.77 \text{ mm}}$$

Applied Calculations: 26 GHz mmWave Mobility Limits

System Parameters

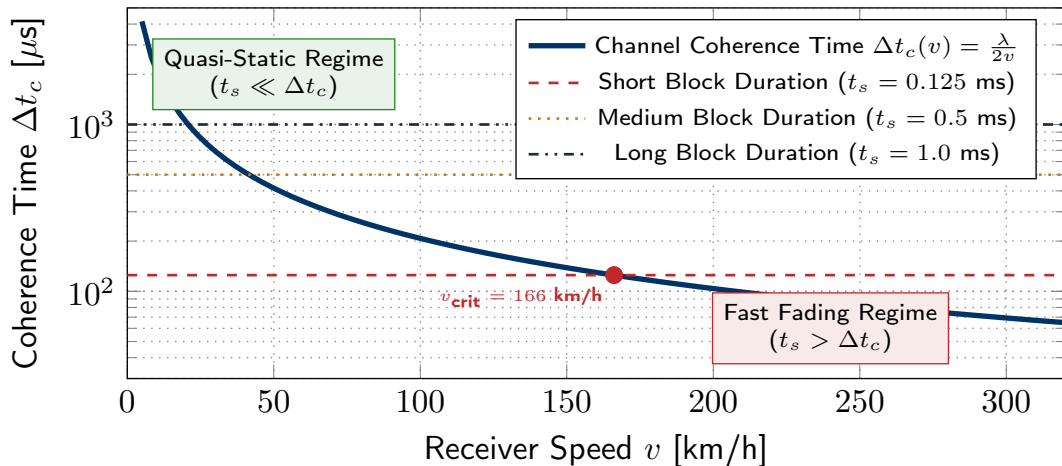
Carrier frequency: $f_c = 26 \text{ GHz} \implies \lambda = \frac{3 \times 10^8}{26 \times 10^9} = 0.01154 \text{ m} = \mathbf{11.54 \text{ mm}}$.

Scenario	$v \text{ [km/h]}$	$v \text{ [m/s]}$	$f_m = \frac{v}{\lambda}$	$\Delta t_c = \frac{1}{2f_m}$	Δd_c	Quasi-Static Status ($t_s = 0.125 \text{ ms}$)
Pedestrian	5	1.39	120.4 Hz	4.15 ms	5.77 mm	Valid: $t_s \ll \Delta t_c$ ($t_s \approx 3\% \Delta t_c$)
City Vehicle	50	13.89	1.204 kHz	415.4 μs	5.77 mm	Marginal: $t_s < \Delta t_c$ ($t_s \approx 30\% \Delta t_c$)
Highway	120	33.33	2.889 kHz	173.1 μs	5.77 mm	Critical: $t_s \sim \Delta t_c$ ($t_s \approx 72\% \Delta t_c$)
TGV	300	83.33	7.222 kHz	69.2 μs	5.77 mm	Breakdown: $t_s > \Delta t_c$ ($t_s \approx 181\% \Delta t_c$)

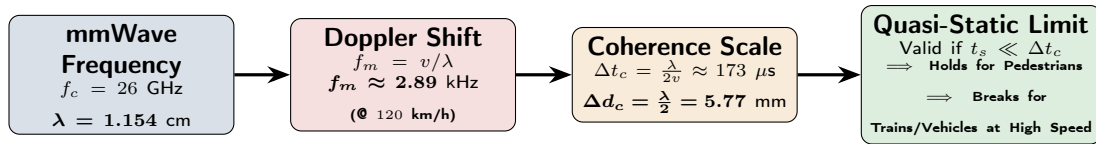
Key Analytical Finding

At 26 GHz, for a typical transmission block $t_s = 0.125 \text{ ms}$, the quasi-static approximation breaks down for speeds above $v \approx 166 \text{ km/h}$, as Δt_c drops below t_s .

Coherence Time vs. Velocity: Breakdown Boundaries



Synthesis: The mmWave Quasi-Static Chain



Key Takeaways

- 1 The channel time variation is governed by the Doppler spread $2f_m = \frac{2v}{\lambda}$, defining the coherence time $\Delta t_c \simeq \frac{\lambda}{2v}$.
- 2 The spatial variation period is strictly $\Delta d_c = \frac{\lambda}{2} = 5.77 \text{ mm}$ at 26 GHz due to constructive and destructive interference of plane waves.
- 3 The quasi-static approximation ($t_s \ll \Delta t_c$) is valid for pedestrians ($\Delta t_c = 4.15 \text{ ms}$), but fails for vehicular speeds ($v > 166 \text{ km/h}$ for $t_s = 125 \mu\text{s}$), transforming the channel into a fast fading environment.